

Equilibrium solvers for the Competition for Risk game: approaches and benchmark

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Abstract

We compare our regret-matching solvers with 21 other methods that compute Nash equilibria of the two-player Competition for Risk (CfR) game with market friction τ and correlated failures ρ [1]. All methods are evaluated with the same metric, the QuasiNashConv with continuous best responses, and the same time limit. We then add a new method, Double Oracle followed by a Newton solve on the atom positions, and we solve the n -player game with friction with the Multiple Oracle algorithm [5]. The best method depends on τ : support generation for $\tau = 0.03$, fictitious play for $\tau = 10^{-4}$ and Newton on the complementarity problem for $\tau = 0$.

1 Game and protocol

Game. Player i picks a risk $r_i \in [0, 1]$ and fails with probability r_i ; the two failures are correlated through a Gaussian copula with correlation ρ , and $\tilde{r}$ is the probability that both fail. With reward $R = 1$ and penalty P , the payoff of player 1 is

$$u_1(r_1, r_2) = (r_2 - \tilde{r}) - r_1 P + (1 - r_1 - r_2 + \tilde{r}) \sigma_\tau(r_1 - r_2),$$

where $\sigma_\tau(x) = 1/(1 + e^{-x/\tau})$ and σ_0 is the step function (value 1/2 on ties). All two-player experiments use $P = 1$, $\rho = 0.5$ and $\tau \in \{0.03, 10^{-4}, 0\}$. For $\tau = 0.03$ the equilibrium has 6 atoms and `analytical.py` computes it exactly; for $\tau = 0$ it is a density on $[0, 0.387]$; for $\tau = 10^{-4}$ no exact solution is available.

Quality. The QuasiNashConv (QNC) of a profile (p_1, p_2) is $\sum_i (\max_{r \in [0,1]} u_i(r, p_{-i}) - u_i(p))$. The maximum over r is continuous, not over the grid of the method: a Sobol scan of 2^{14} points, the opponent's atoms and the points at $\pm 10^{-12}$ of them (the supremum is a one-sided limit when $\tau = 0$), then Brent refinement of the 8 best points. The exact equilibrium at $\tau = 0.03$ gets QNC $1.1 \cdot 10^{-16}$. At $\tau = 0$, the analytical density cut into 8192 equal-mass atoms gets $1.2 \cdot 10^{-4}$.

Time. Every solve runs in its own process with one thread (BLAS, numba, Gurobi, HiGHS, XLA), on an Apple M4 Max on AC power, with at most 10 solves in parallel (fewer than the 12 performance cores). A solve that has not finished after 120s returns its best point if the solver allows it; a hard timeout kills it at 240s. Time is the construction of the payoff matrix plus the solve. Grid methods use the grid $\{0, 1/(n-1), \dots, 1\}$ for both players, with $n = 16, 32, \dots, 4096$. For a method, the grid sizes run in increasing order and stop at the first failure.

Iterations. An iterative method runs once per n and saves its strategy after chunks of 1, 1, 2, 4, 8, $\dots$ iterations, until 120s. Each snapshot is one point $(n, \text{iterations})$. For every method we keep the lower envelope over all its points: the best QNC that is reachable within a given time. One iteration of every iterative method costs $O(n^2)$: for the sampled methods (RM, fictitious play) one “iteration” is a round of n updates.

2 Methods

Method	Description
RM, CFR (ours)	Regret matching with sampled updates, and its full-information version (CFR), average strategies [6, 7, 1]. Also with shifted grids: the two players use interleaved grids, as in the paper.
Lemke–Howson (Gambit, numpy)	Complementary pivoting on the bimatrix game [3]: Gambit in floating point [19], and a numpy tableau with the lexicographic ratio test, after [20].
Logit QRE path (Gambit)	Path of logit quantal response equilibria as $\lambda \rightarrow \infty$ [12, 13].
Mangasarian–Stone (Gurobi)	QP The quadratic program of [2]: $\min \alpha + \beta - p^\top (A+B)q$ s.t. $Aq \leq \alpha \mathbf{1}, B^\top p \leq \beta \mathbf{1}, p, q$ in the simplex. Its global minima (value 0) are the Nash equilibria. Nonconvex, solved globally by Gurobi.
Nash-gap QP, $2n$ bilinear (Gurobi, SCIP)	The same program as [2] with $w = Aq$ and $s = B^\top p$ as variables: the objective $\alpha + \beta - p^\top w - q^\top s$ has $2n$ bilinear terms instead of n^2 .
FB-NCP Newton	Symmetric equilibrium x as a complementarity problem, $x \geq 0, v\mathbf{1} - Ax \geq 0, x^\top (v\mathbf{1} - Ax) = 0$, with the Fischer–Burmeister function [14], smoothing and damped Newton steps.
Ipopt multi-start	The symmetric Nash-gap program, local interior-point solves [15] from 5 starts.
Double Oracle, grid / continuous	Restricted game on a growing support; each iteration adds up to 8 profitable local maxima of the deviation payoff per player [4]. Best responses on the grid, or over $[0, 1]$ (Sobol scan of n points and Brent refinement).
Double Oracle + Newton	New: Section 5.
Fictitious play, Hedge, replicator	Simultaneous fictitious play [8, 9], multiplicative weights with step $\eta_t = \sqrt{8 \log n/t}$ [10], discrete replicator dynamics [11].
nashopt: MILP (HiGHS, Gurobi), Lemke, GNEP, extra-gradient, DR-DAQP	The bimatrix game as a linear-quadratic generalized Nash problem in nashopt [18]: big-M mixed-integer program of the KKT conditions (the approach of [16]), Lemke’s algorithm, least squares on the Fischer–Burmeister KKT residual, extra-gradient [17], Douglas–Rachford with active set.

Table 1: The 24 methods. The solvers that need a monotone game (nashopt **qp_gnep**, **op_extrap**) do not apply: the pseudo-gradient of this game is not monotone.

3 Results

Table 2 gives the best QNC of each method over the whole run, and Table 3 the best QNC within 10 s. Figure 1 shows the lower envelopes. Full tables, with the n and iteration counts of each point and the first failing grid size of each method, are in **bench/tau=*/RESULTS.md** and **bench/SUMMARY.md**.

Method	$\tau = 0.03$	$\tau = 10^{-4}$	$\tau = 0$
Double Oracle + Newton on atoms	$6.7 \cdot 10^{-16}$ (2.8 s)	$8.2 \cdot 10^{-3}$ (90 s)	—
Double Oracle, continuous BR	$2.9 \cdot 10^{-11}$ (2.5 s)	$6.7 \cdot 10^{-3}$ (120 s)	$8.2 \cdot 10^{-3}$ (31 s)
Double Oracle, grid BR	$9.0 \cdot 10^{-8}$ (2.5 s)	$5.1 \cdot 10^{-3}$ (15 s)	$6.8 \cdot 10^{-3}$ (15 s)
FB-NCP Newton, symmetric	$9.0 \cdot 10^{-8}$ (9.8 s)	$8.5 \cdot 10^{-5}$ (128 s)	$8.4 \cdot 10^{-4}$ (128 s)
Lemke-Howson (numpy tableau)	$9.0 \cdot 10^{-8}$ (34 s)	$2.1 \cdot 10^{-2}$ (0.44 s)	$2.3 \cdot 10^{-2}$ (0.44 s)
Fictitious play	$1.1 \cdot 10^{-7}$ (88 s)	$6.7 \cdot 10^{-5}$ (84 s)	$9.2 \cdot 10^{-4}$ (51 s)
nashopt Lemke	$1.7 \cdot 10^{-6}$ (46 s)	$6.8 \cdot 10^{-3}$ (17 s)	$8.7 \cdot 10^{-3}$ (13 s)
Nash-gap QP, 2n bilinear (Gurobi)	$3.7 \cdot 10^{-6}$ (121 s)	$6.9 \cdot 10^{-4}$ (121 s)	$2.0 \cdot 10^{-3}$ (121 s)
Ipopt multi-start	$4.7 \cdot 10^{-6}$ (97 s)	$7.5 \cdot 10^{-4}$ (44 s)	$2.0 \cdot 10^{-3}$ (50 s)
Nash-gap QP, 2n bilinear (SCIP)	$5.2 \cdot 10^{-6}$ (124 s)	$7.2 \cdot 10^{-4}$ (21 s)	$2.0 \cdot 10^{-3}$ (29 s)
Replicator dynamics	$8.8 \cdot 10^{-6}$ (35 s)	$2.0 \cdot 10^{-3}$ (93 s)	$3.4 \cdot 10^{-3}$ (68 s)
Mangasarian-Stone QP (Gurobi)	$1.0 \cdot 10^{-5}$ (120 s)	$6.8 \cdot 10^{-3}$ (120 s)	$8.7 \cdot 10^{-3}$ (120 s)
CFR (ours)	$2.2 \cdot 10^{-5}$ (66 s)	$2.1 \cdot 10^{-4}$ (120 s)	$1.3 \cdot 10^{-3}$ (74 s)
Logit QRE path (Gambit)	$3.6 \cdot 10^{-5}$ (74 s)	$6.8 \cdot 10^{-3}$ (22 s)	$8.7 \cdot 10^{-3}$ (24 s)
Multiplicative weights	$2.8 \cdot 10^{-4}$ (91 s)	$6.0 \cdot 10^{-4}$ (101 s)	$1.5 \cdot 10^{-3}$ (128 s)
nashopt extragradient	$3.1 \cdot 10^{-4}$ (74 s)	$1.1 \cdot 10^{-2}$ (68 s)	$1.3 \cdot 10^{-2}$ (81 s)
RM (ours), shifted grids	$6.2 \cdot 10^{-4}$ (102 s)	$7.3 \cdot 10^{-4}$ (103 s)	$1.6 \cdot 10^{-3}$ (94 s)
RM (ours)	$7.4 \cdot 10^{-4}$ (12 s)	$7.8 \cdot 10^{-4}$ (101 s)	$1.6 \cdot 10^{-3}$ (93 s)
nashopt MILP (Gurobi)	$7.5 \cdot 10^{-4}$ (13 s)	$4.2 \cdot 10^{-2}$ (0.87 s)	$4.4 \cdot 10^{-2}$ (0.73 s)
nashopt MILP (HiGHS)	$7.8 \cdot 10^{-4}$ (117 s)	$6.8 \cdot 10^{-3}$ (37 s)	$8.7 \cdot 10^{-3}$ (36 s)
Lemke-Howson (Gambit)	$7.8 \cdot 10^{-4}$ (0.057 s)	$6.8 \cdot 10^{-3}$ (204 s)	$8.7 \cdot 10^{-3}$ (198 s)
CFR (ours), shifted grids	$8.5 \cdot 10^{-4}$ (6.7 s)	$1.1 \cdot 10^{-3}$ (122 s)	$2.0 \cdot 10^{-3}$ (60 s)
nashopt GNEP (FB least squares)	$3.1 \cdot 10^{-3}$ (0.93 s)	$4.2 \cdot 10^{-2}$ (5.3 s)	$4.4 \cdot 10^{-2}$ (4.1 s)
nashopt DR-DAQP	$2.6 \cdot 10^{-2}$ (0.00088 s)	$7.2 \cdot 10^{-4}$ (113 s)	$2.0 \cdot 10^{-3}$ (96 s)

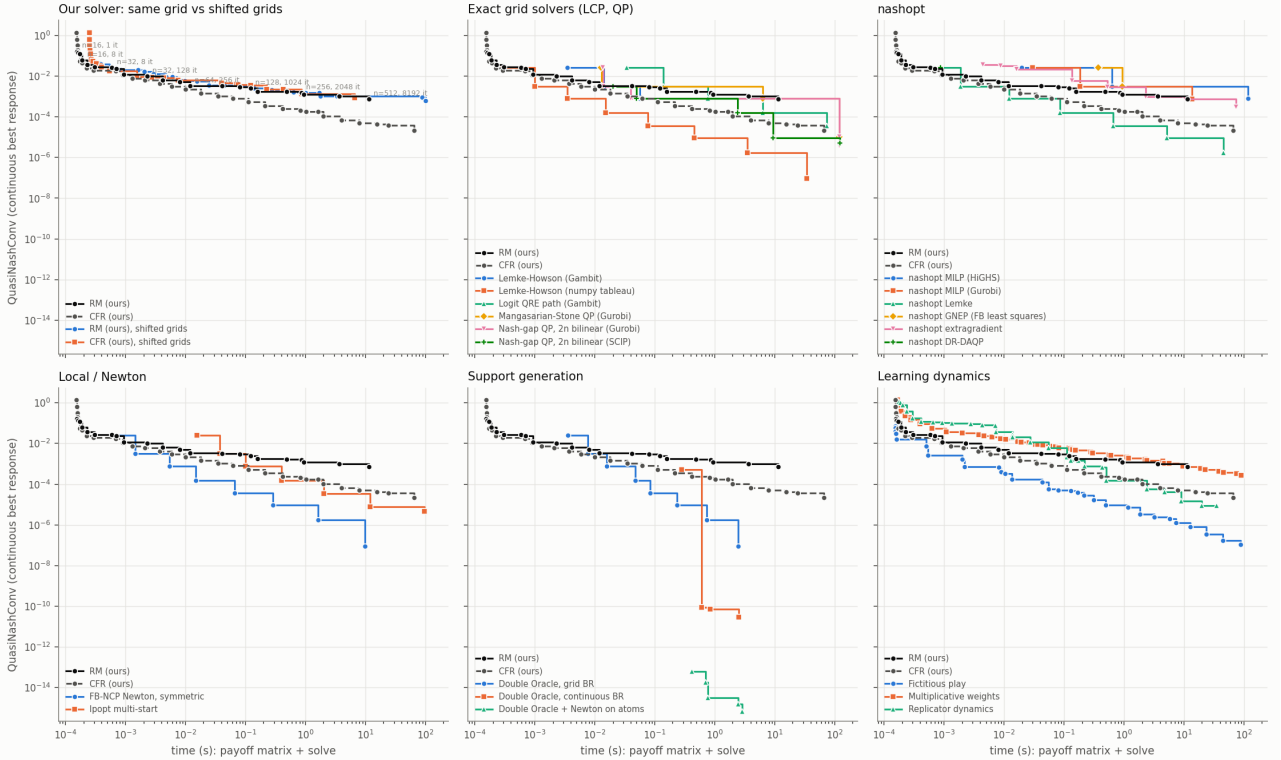
Table 2: Best QNC over the whole run (time in brackets), sorted by the result at $\tau = 0.03$. “—”: no result (failure at the smallest grids, or not applicable).

Method	$\tau = 0.03$	$\tau = 10^{-4}$	$\tau = 0$
Double Oracle + Newton on atoms	$6.7 \cdot 10^{-16}$	—	—
Double Oracle, continuous BR	$2.9 \cdot 10^{-11}$	—	$1.1 \cdot 10^{-2}$
Double Oracle, grid BR	$9.0 \cdot 10^{-8}$	$6.8 \cdot 10^{-3}$	$8.7 \cdot 10^{-3}$
FB-NCP Newton, symmetric	$9.0 \cdot 10^{-8}$	$7.2 \cdot 10^{-4}$	$2.0 \cdot 10^{-3}$
Lemke-Howson (numpy tableau)	$1.7 \cdot 10^{-6}$	$2.1 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$
Fictitious play	$1.3 \cdot 10^{-6}$	$5.5 \cdot 10^{-4}$	$1.3 \cdot 10^{-3}$
nashopt Lemke	$9.3 \cdot 10^{-6}$	$2.1 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$
Nash-gap QP, 2n bilinear (Gurobi)	$7.8 \cdot 10^{-4}$	$1.2 \cdot 10^{-1}$	$1.3 \cdot 10^{-1}$
Ipopt multi-start	$3.5 \cdot 10^{-5}$	$6.8 \cdot 10^{-3}$	$8.7 \cdot 10^{-3}$
Nash-gap QP, 2n bilinear (SCIP)	$9.2 \cdot 10^{-6}$	$5.1 \cdot 10^{-3}$	$6.8 \cdot 10^{-3}$
Replicator dynamics	$1.5 \cdot 10^{-5}$	$6.5 \cdot 10^{-3}$	$7.9 \cdot 10^{-3}$
Mangasarian-Stone QP (Gurobi)	$7.8 \cdot 10^{-4}$	$1.2 \cdot 10^{-1}$	$4.4 \cdot 10^{-2}$
CFR (ours)	$5.1 \cdot 10^{-5}$	$8.6 \cdot 10^{-4}$	$2.1 \cdot 10^{-3}$
Logit QRE path (Gambit)	$1.6 \cdot 10^{-4}$	$2.1 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$
Multiplicative weights	$8.0 \cdot 10^{-4}$	$1.7 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$
nashopt extragradient	$9.5 \cdot 10^{-4}$	$2.8 \cdot 10^{-2}$	$3.1 \cdot 10^{-2}$
RM (ours), shifted grids	$1.0 \cdot 10^{-3}$	$2.0 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$
RM (ours)	$1.0 \cdot 10^{-3}$	$2.1 \cdot 10^{-3}$	$3.1 \cdot 10^{-3}$
nashopt MILP (Gurobi)	$3.1 \cdot 10^{-3}$	$4.2 \cdot 10^{-2}$	$4.4 \cdot 10^{-2}$
nashopt MILP (HiGHS)	$3.1 \cdot 10^{-3}$	$4.2 \cdot 10^{-2}$	$4.4 \cdot 10^{-2}$
Lemke-Howson (Gambit)	$7.8 \cdot 10^{-4}$	$2.1 \cdot 10^{-2}$	$2.3 \cdot 10^{-2}$
CFR (ours), shifted grids	$8.5 \cdot 10^{-4}$	$1.9 \cdot 10^{-3}$	$2.6 \cdot 10^{-3}$
nashopt GNEP (FB least squares)	$3.1 \cdot 10^{-3}$	$4.2 \cdot 10^{-2}$	$4.4 \cdot 10^{-2}$
nashopt DR-DAQP	$2.6 \cdot 10^{-2}$	$6.8 \cdot 10^{-3}$	$8.7 \cdot 10^{-3}$

Table 3: Best QNC within 10s.

Observations. For $\tau = 0.03$, the continuous methods are far ahead: Double Oracle reaches $3 \cdot 10^{-11}$ and Double Oracle + Newton the exact equilibrium, in about 3s. The grid methods cannot do better than the exact equilibrium of the grid (QNC $9 \cdot 10^{-8}$ at $n = 2048$); the fastest of them are grid Double Oracle and FB-NCP Newton. CFR reaches $2 \cdot 10^{-5}$ and RM $7 \cdot 10^{-4}$.

Accuracy vs running time, 2-player CFR game ($P = 1$, $\tau = 0.03$, $\rho = 0.5$), grids $n = 16 \dots 4096$



Accuracy vs running time, 2-player CFR game ($P = 1$, $\tau = 0.0001$, $\rho = 0.5$), grids $n = 16 \dots 4096$

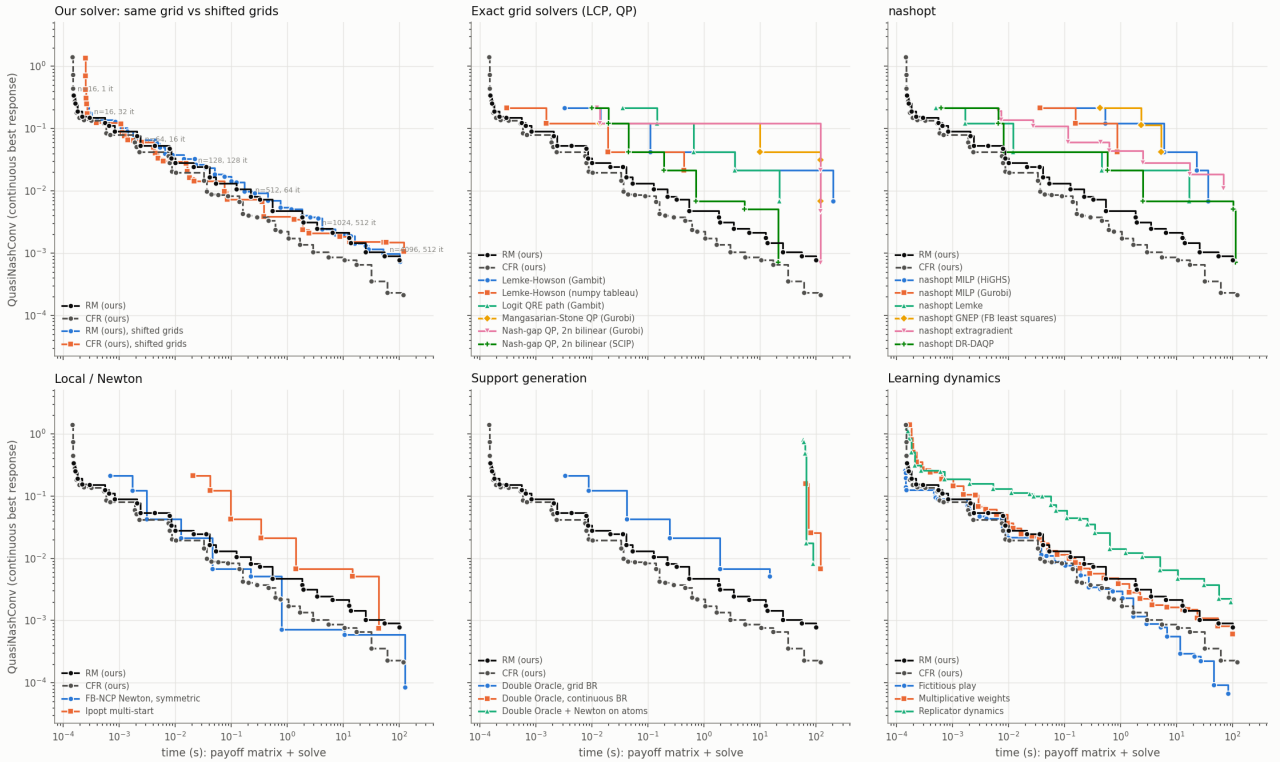


Figure 1: QNC against time, lower envelope over $(n, \text{iterations})$, for $\tau = 0.03$ (top) and $\tau = 10^{-4}$ (bottom). The figure for $\tau = 0$ is [plots/benchmark_accuracy_time_tau=0.svg](#).

For $\tau = 10^{-4}$ and $\tau = 0$ the equilibrium is, at the resolution of the grids, a density on $[0, 0.387]$. Support generation then adds about one point per iteration and falls behind. Fictitious play is the best method at $\tau = 10^{-4}$ ($6.7 \cdot 10^{-5}$), FB-NCP Newton at $\tau = 0$ ($8.4 \cdot 10^{-4}$); CFR is 3 times worse at $\tau = 10^{-4}$ and 1.5 times worse at $\tau = 0$. The Lemke–Howson paths become very long on these nearly degenerate games (16 000 pivots at $n = 128$ for $\tau = 0$, against 288 for $\tau = 0.03$): no Lemke–Howson implementation gives a better point than its result at $n = 256$.

The two formulations of the Mangasarian–Stone program find a solution close to an equilibrium within seconds for $n \leq 64$, but they cannot prove optimality (the lower bound stays negative) and use the full time limit from $n = 128$. The $2n$ -bilinear version is better than the dense one. The mixed-integer programs (nashopt) find no feasible point within 120 s from $n = 128$ with Gurobi, and from $n = 128$ or $n = 512$ with HiGHS.

4 Supports

Table 4 compares the Double Oracle solutions at $\tau = 0.03$ with the exact equilibrium. Double Oracle finds the 6 atoms but it never moves a point: when a best response lies next to a true atom, the restricted equilibrium mixes two close points, whose weighted centre is the atom. Most of the 16–70 points that it adds end with weight 0.

DO	n	QNC	points added	weight > 0	points per atom	position error	weight error
continuous	64	$3.8 \cdot 10^{-10}$	65	9	1,2,1,2,2,1	$1.5 \cdot 10^{-5}$	$3.8 \cdot 10^{-5}$
continuous	256	$7.2 \cdot 10^{-11}$	61	9	1,1,2,1,2,2	$3.1 \cdot 10^{-6}$	$4.9 \cdot 10^{-6}$
continuous	1024	$1.1 \cdot 10^{-10}$	68	11	1,2,2,2,2,2	$7.3 \cdot 10^{-6}$	$1.5 \cdot 10^{-5}$
continuous	4096	$8.1 \cdot 10^{-11}$	70	9	1,2,1,2,2,1	$1.4 \cdot 10^{-5}$	$3.4 \cdot 10^{-5}$
grid	64	$7.8 \cdot 10^{-4}$	16	9	1,2,1,1,2,2	$2.5 \cdot 10^{-2}$	$4.4 \cdot 10^{-2}$
grid	256	$3.6 \cdot 10^{-5}$	26	10	1,1,2,2,2,2	$3.8 \cdot 10^{-3}$	$5.1 \cdot 10^{-3}$
grid	1024	$1.7 \cdot 10^{-6}$	36	9	1,2,1,2,2,1	$6.7 \cdot 10^{-4}$	$1.2 \cdot 10^{-3}$
grid	4096	$1.5 \cdot 10^{-7}$	45	9	1,1,2,1,2,2	$2.1 \cdot 10^{-4}$	$3.0 \cdot 10^{-4}$

Table 4: Double Oracle against the exact 6 atoms, $\tau = 0.03$. Each support point is assigned to the nearest exact atom; errors are for the weighted centres and the summed weights.

Method	n	QNC	weight $> 10^{-9}$	99.9% of the mass	range
Fictitious play	4096	$6.7 \cdot 10^{-5}$	1584	1582	[0.000, 0.387]
CFR (ours)	2048	$2.1 \cdot 10^{-4}$	2048	791	[0.000, 0.386]
Double Oracle, grid BR	512	$5.1 \cdot 10^{-3}$	169	155	[0.000, 0.386]
Double Oracle, continuous BR	512	$6.7 \cdot 10^{-3}$	299	292	[0.000, 0.386]

Table 5: Supports at $\tau = 10^{-4}$ (best result of each method).

At $\tau = 10^{-4}$ (Table 5) fictitious play and CFR put their mass on the same interval, which covers almost all grid points of $[0, 0.387]$; fictitious play gives zero weight to the other points, and CFR keeps a small weight on every point (the first, uniform iterations of the average strategy).

5 Double Oracle followed by Newton on the atoms

`analytical.py` also grows a support, but it moves the atoms: for a support $x_1 < \dots < x_m$ it solves, in positions, weights and value v ,

$$\sum_j u(x_i, x_j) w_j = v \ (\forall i), \quad \sum_j \partial_1 u(x_i, x_j) w_j = 0 \ (x_i > 0), \quad \sum_j w_j = 1.$$

The first-order condition places each interior atom at a local maximum of the deviation payoff; Double Oracle has no such condition. The square system needs exactly the right number of atoms, so we start from the Double Oracle solution and solve it with a Newton method that handles pairs: two atoms closer than $\tau/10$ (the length scale of the payoff) merge into one, an atom whose weight becomes non-positive leaves the support, the closest pair merges when the residual falls by less than 10% in 20 iterations, and the steps are damped (Levenberg–Marquardt). The Jacobian is exact: $\partial_2 u$ follows from $u(a, b) + u(b, a) = 1 - \tilde{r}(a, b) - P(a + b)$. Among the start, the states before each forced merge and the final state, we keep the one with the smallest continuous best-response gap, measured on its own Sobol scan: on the scan of the Double Oracle oracle, the start always looks converged.

τ	start	QNC start	time (s)	QNC after	Newton (s)	atoms	merges
0.03	Double Oracle	$8.1 \cdot 10^{-11}$	8.7	$3.2 \cdot 10^{-15}$	0.13	6	3+0
0.03	fictitious play	$8.5 \cdot 10^{-5}$	43	$1.1 \cdot 10^{-14}$	0.16	6	7+0
0.01	Double Oracle	$8.9 \cdot 10^{-7}$	14	$5.8 \cdot 10^{-9}$	26	32	12+31
0.01	fictitious play	$6.3 \cdot 10^{-5}$	43	$2.6 \cdot 10^{-8}$	117	45	164+64
0.001	Double Oracle	$2.5 \cdot 10^{-6}$	305	$6.0 \cdot 10^{-7}$	647	460	34+10
0.001	fictitious play	$2.1 \cdot 10^{-5}$	45	$1.5 \cdot 10^{-5}$	646	765	14+4
0.0001	Double Oracle	$6.1 \cdot 10^{-1}$	303	$6.1 \cdot 10^{-1}$	1.1	76	6+0
0.0001	fictitious play	$9.3 \cdot 10^{-5}$	51	$9.3 \cdot 10^{-5}$	658	1584	0+1

Table 6: Newton on the atoms after Double Oracle (300 s) or fictitious play (60 s), single thread. Merges: collisions + forced merges.

Table 6 gives the results. For $\tau = 0.03$ both starts give the exact equilibrium in 7 Newton steps. For $\tau = 10^{-2}$ the error falls by a factor of about 150. For $\tau \leq 10^{-3}$ the support has hundreds of atoms, each Newton step costs seconds, and the gain is small or zero. In the benchmark (Table 2) the method uses half of the time limit for Double Oracle and half for Newton; it does not apply to $\tau = 0$ (the payoff is not smooth).

6 More than two players

Game. With n players and independent failures ($\rho = 0$), among the players that do not fail, player i wins R with probability $e^{r_i/\tau} / \sum_j e^{r_j/\tau}$; for $n = 2$ this is the friction model above. If the $n - 1$ opponents play the mixed strategy (a_k, w_k) , the identity $1/x = \int_0^\infty e^{-tx} dt$ with $t = e^s$ gives

$$U(r) = -Pr + (1 - r) \int e^{s-e^s} G(s, r)^{n-1} ds, \quad G(s, r) = \sum_k w_k [a_k + (1 - a_k) \exp(-e^{s+(a_k-r)/\tau})],$$

one integral whatever n . This formula agrees with `game.py` for $n = 2$ to $3 \cdot 10^{-14}$ and with a Monte Carlo simulation for $n = 3, 4$.

Multiple Oracle. We apply Algorithm 1 of Kroupa and Votroubek [5]. The game is symmetric, so the players share one strategy set and one best response; the restricted game is the symmetric complementarity problem in the weights, solved by Fischer–Burmeister Newton followed by Newton on the support. Its payoffs are polynomials of degree $n - 1$ in the weights, so each restricted solve is a nonlinear problem, but it has k unknowns and not a payoff table of k^n entries.

	Double Oracle [4]	Multiple Oracle [5]
Games	two players, zero-sum, finite	n players, general-sum, compact action sets, continuous payoffs
Restricted game	linear program	n -player Nash equilibrium (PATH in [5]; here a symmetric complementarity problem)
Best response	over a finite set	global optimization over the continuous set
Stop	finite: the action sets are finite	finite for $\varepsilon > 0$; for $\varepsilon = 0$ every limit point is an equilibrium

Table 7: Standard Double Oracle and Multiple Oracle.

n	NashConv	check	time (s)	iterations	$ X $	weight > 0	atoms	mass at 0
2	$2.9 \cdot 10^{-7}$	$2.9 \cdot 10^{-7}$	417	56	56	12	6	0.106
3	$1.4 \cdot 10^{-7}$	$1.4 \cdot 10^{-7}$	420	66	66	18	9	0.165
5	$2.2 \cdot 10^{-9}$	$1.9 \cdot 10^{-7}$	404	71	71	13	8	0.415
10	$8.5 \cdot 10^{-11}$	$1.2 \cdot 10^{-10}$	293	66	66	11	7	0.720

Table 8: Multiple Oracle, $\tau = 0.03$, $P = 1$, $\rho = 0$, 400 s limit, 2 threads. Check: NashConv with a finer best response. Atoms: groups of support points less than $2 \cdot 10^{-3}$ apart.

Table 8 gives the results. The number of atoms stays small (6 to 9) and the mass at zero risk grows with n . For $n = 5$ the oracle reports $2.2 \cdot 10^{-9}$ but the finer check gives $1.9 \cdot 10^{-7}$: the oracle scan (2^{12} points) misses the best deviation. Removing the zero-weight strategies after each restricted solve (not in [5]) makes the

algorithm cycle: for $n = 2$, after 100 iterations the set still has 2 strategies and NashConv repeats the sequence 0.56, 0.43, 0.32, 0.24, 0.20. The restricted solves are much slower than in the two-player case, where the restricted game is bilinear and its support system linear: on the same game ($n = 2$, $\rho = 0$), the two-player Double Oracle reaches $1.6 \cdot 10^{-10}$ in 1.7 s, against $2.9 \cdot 10^{-7}$ in 417 s for the Multiple Oracle (`bench/multiplayer/checks.json`).

7 Limits

All two-player results are for $P = 1$ and $\rho = 0.5$, and the n -player results for $\tau = 0.03$ and $\rho = 0$. The time limit (120 s) favours methods that give a usable point early. Gurobi used a full license (WLS); an earlier size-limited license stopped the Gurobi methods at 2000 variables.

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